#1 · 12.7% of marks

Ch 9 — Derivatives Analysis & Valuation

Tested every attempt: strangle / straddle payoffs, index-futures beta hedge, Black-Scholes, real options (risk-neutral), futures + options combos. Theory: real vs financial options.

① Users & Forward vs Futures (SM)

Users: *corporations — hedge currency & inventory risk · individual investors — speculation, hedging, yield enhancement · institutional investors — hedging asset allocation, yield enhancement, arbitrage · dealers — hedging, position taking, exploiting inefficiencies, dealer spreads.*

Feature	Forward	Futures
Trading	Personal basis / telephone	Competitive arena
Size	Individually tailored	Standardised
Exchange	OTC	Organised exchange
Settlement	On agreed date	Daily via clearing house
Delivery date	As agreed; actual delivery	Cyclical; delivery rarely takes place
Transaction cost	Bid-ask spread	Brokerage on buy & sell
Marking to market	No	Yes — daily in margin account
Margins	Not required	Required
Credit risk	Borne by each party	Not a concern (clearing house)

Basis = Spot − Futures. *Normal (**contango**) market: futures > spot (full cost of carry) → basis negative. **Backwardation / inverted**: spot > futures (factors other than carry) → basis positive. Basis → 0 at expiry = **convergence**.*

② Futures pricing — cost of carry

Situation	Formula
Basic (discrete)	$F = S + \text{cost of carry} - \text{dividend} = S(1+r)^t - D$
Continuous compounding	$F = S \cdot e^{rt}$
Known dividend	$F = (S - \text{PV of dividend}) \cdot e^{rt}$
Index with dividend yield y	$F = S \cdot e^{(r-y)t}$
Commodity with storage cost	$F = (S + \text{PV storage}) \cdot e^{rt}$ (convenience yield reduces F)

Actual F > Theoretical F

*Futures overpriced → **cash-and-carry**: borrow, **buy spot, sell futures**; deliver at expiry.*

Actual F < Theoretical F

*Futures underpriced → **reverse cash-and-carry**: **short-sell spot, invest, buy futures**.*

*SM: Future price = Spot + carrying cost − returns (dividends etc.). RTP Jan 2026: S = 125, r 8%, D = 4 → F = 125 + 10 − 4 = **₹131**; contract (1,000 shares) ₹1,31,000. Price falls 6% → S = 117.50 → F = 122.90 → long loses ₹8,100.*

REPEAT R17

③ Margins & mark-to-market

- → **Initial margin = μ + 3σ** of daily changes in contract value (SM illustration: μ ₹10,000 + 3 × ₹2,000 = ₹16,000; maintenance 75% = ₹12,000)
- → **Maintenance margin** (usually lower); balance falls below it → **margin call** to restore to **initial** margin
- → Daily MTM: long gains when price rises; losses debited to margin account. Cumulative MTM = (final − entry price) × lot

④ Beta hedge — formulas

No. of contracts to fully hedge = (Portfolio value × β_p) ÷ (Futures price × lot size)

Change beta from β_1 to β_2: contracts = ($\beta_2 - \beta_1$) × Portfolio value ÷ (Futures price × lot) — negative = **sell**, positive = **buy**

Long stock + short stock: hedge each leg separately (sell futures for long leg, buy for short leg) or net the beta-weighted values

Example	Working
Portfolio ₹1 cr, β 1.2; Nifty futures 22,000; lot 75	1,00,00,000 × 1.2 ÷ (22,000 × 75) = 7.27 → **sell 7 contracts**
Reduce β 1.2 → 0.6 (same data)	(0.6 − 1.2) × 1 cr ÷ 16,50,000 = −3.64 → sell **4**
RTP Jan 2026: portfolio 9,94,450, 5 Nifty contracts sold at 8,767.07 × 25 for full hedge	Implied β = 5 × 8,767.07 × 25 ÷ 9,94,450 = **1.102**; 1.102 = 0.9 × 1.10 + 0.1 × β_{10} → β_{10} = **1.12**

⚠ TRAP

Use the **futures** price (not spot index) in the denominator. Lot size is whatever the question gives — don't assume. Profit on hedge = (sell price − buy-back price) × lot × contracts; overall P/L = cash position change + futures P/L.

⑤ Options vs single stock futures (SM 6.7)

Point	Options	Stock futures
Nature	Buyer has right, not obligation	Long must square off at / before expiry
Movement of market	Premium moves with **delta**; may move independently of stock	Follows the underlying much more closely
Price of investing	Premium = **sunk cost**; lost if not ITM at expiry	Initial margin + maintenance (MTM) margin

7 Basics

	Call (right to BUY at K)	Put (right to SELL at K)
In the money	$S > K$	$S < K$
At the money	$S = K$	$S = K$
Out of the money	$S < K$	$S > K$
Intrinsic value	$\max(S - K, 0)$	$\max(K - S, 0)$
Time value	Premium − intrinsic value (all premium is time value if OTM / ATM)	
Buyer's net pay-off	$\max(S_T - K, 0)$ − premium	$\max(K - S_T, 0)$ − premium
Writer's pay-off	Exact mirror image of buyer (zero-sum)	

⚠ TRAP

Exercise decision ignores premium (sunk) — exercise whenever ITM, even if net result is still a loss.

Premium & brokerage are per share × **lot**. European = exercise only on expiry; American = any time.

8 Strategies & break-evens

exam aid — not described in SM; PYQs test the pay-offs

Strategy	Position	Break-even / view
Long straddle	Buy call + buy put, **same K**	K ± total premium; expects big move either way
Long strangle	Buy OTM call (K_c) + OTM put (K_p), $K_c > K_p$	Upper = K_c + total premium; lower = K_p − total premium
Strip	1 call + 2 puts (same K)	Bearish volatility bet
Strap	2 calls + 1 put	Bullish volatility bet
Bull spread	Buy low-K call, sell high-K call	Limited gain / loss; moderately bullish
Bear spread	Buy high-K put, sell low-K put (or calls reversed)	Moderately bearish
Butterfly	Buy 1 low-K + 1 high-K call, sell 2 middle-K calls	Expects price to stay near middle K
Covered call	Long stock / long futures + **sell call**	Earns premium; caps upside (Jan 2026 Q3(a))
Protective put	Long stock + buy put	Floor on loss (insurance)

Long strangle: call 620 (premium 25) + put 480 (premium 15); total premium 40; BE 660 / 440				
Expiry price	700	550	400	660 / 440
Net P/L per share	+40	**−40** (max loss)	+40	0

⑨ Binomial — one step (three methods, same answer)

$S = 100$; up 120, down 80; $K = 100$; r_f 10% for the period.

Method	Working	Call
Risk-neutral	$p = (S(1+r) - S_d) \div (S_u - S_d) = (110 - 80)/40 = 0.75$; $C = (0.75 \times 20 + 0.25 \times 0) \div 1.10$	**13.64**
Delta / replicating	$\Delta = (C_u - C_d)/(S_u - S_d) = 20/40 = 0.5$; borrow $B = (\Delta S_d - C_d)/1.1 = 36.36$; $C = \Delta S - B$	**13.64**
Put (parity)	$P = C - S + K/(1 + r) = 13.64 - 100 + 90.91$	**4.55**

Two-step: $u = 1.1$, $d = 0.9$, r 5% per step → $p = 0.75$; terminal call values 21, 0, 0 → $C = 0.75^2 \times 21 \div 1.05^2$ = **10.71**. Work backwards node by node (needed for American puts: compare early exercise at each node).

⑩ Black-Scholes (European, non-dividend)

$$C = S \cdot N(d_1) - K \cdot e^{-rt} \cdot N(d_2)$$

$d_1 = [\ln(S/K) + (r + \sigma^2/2)t] \div \sigma\sqrt{t}$ · $d_2 = d_1 - \sigma\sqrt{t}$

$P = C - S + K \cdot e^{-rt}$ (put-call parity) or $P = K \cdot e^{-rt} N(-d_2) - S \cdot N(-d_1)$

$N(d_1)$ = hedge ratio / delta · **$N(d_2)$ = probability call finishes ITM** · $N(-d) = 1 - N(d)$ · interpolate N() from the table

May 2026 Q6(b): S 120, K 115, 50 days, r 8%, σ 30%, d_1 0.5375, e^{-rt} 0.9891 → $N(d_1)$ **0.7045** (hedge ratio ≈ 70%); $d_2 = 0.5375 - 0.30\sqrt{(50/365)}$ = **0.4264**; $N(d_2)$ **0.6651** (≈ 67%); $C = 84.54 - 75.65$ = **₹8.89**; $P = -120 + 8.89 + 113.75$ = **₹2.63** (ICAI). Verified by code: C 8.89, P 2.64 (rounding).

⑪ BSM assumptions, factors & Greeks (SM)

Assumptions (7): European options · no transaction costs · short-term interest rates known & constant · stocks pay no dividend · stock price follows a random walk · returns normally distributed · variance of return constant over option life.

Factor affecting value (SM)	Effect
(a) Price of underlying	Price ↑ → calls ↑, puts ↓
(b) Time till expiry	Option may become worthless at expiry — time value decays
(c) Volatility	Statistical / historical volatility (SV); higher volatility → higher value
(d) Interest rate	Higher rates → lower PV of exercise price (call ↑, put ↓)

Greek	SM definition
Delta	Degree to which option price moves for a small change in underlying price
Gamma	How fast delta changes — "delta of the delta"; keep small for delta hedging; adjusted only with options
Theta	Change in option price for a **one-day decrease** in time to expiry — time decay
Rho	Change in option price for a **one percentage point** change in risk-free rate
Vega	Sensitivity of option value to change in volatility; used for hedging

REPEAT R13

12 Real options — types (SM)

Option	SM meaning	Like
Growth	Strategic investments: define competitive position, gain knowledge about project profitability, give manufacturing flexibility	European call
Abandonment	Exercised if value from project's assets > PV of continuing the project	American put
Timing	Third choice besides accept / reject: **delay** the decision (when to invest)	American call

Valuation methods = same as financial options (binomial, risk-neutral, BSM). **Expanded NPV = static NPV + option value.**

13 Real vs financial options (SM table — Jan 2026, 4 marks)

Basis	Financial options	Real options
Underlying	Assets normally traded — shares, bonds, commodities	Projects — not traded
Pay-off	Specified in contract → fixed	Estimated from project cash flows → can vary
Exercise period	Short — maximum up to 1 year	Starts from end of 1st year; longer
Approach	Traded → **"Priced"**	Used for decisions → **"Valued"**

14 Risk-neutral real-option numbers (app-computed)

Case	Working
Abandonment: PV now 100; next year 140 or 70; salvage 90; r_f 10%	$p = (110 - 70)/(140 - 70) = 0.5714$; value = 0.4286 × 20 ÷ 1.1 = **₹7.79**
Option to wait: invest 100; PV now 105; next year 130 or 85; r_f 8%	$p = 0.6311$; value = 0.6311 × 30 ÷ 1.08 = **₹17.53** > NPV 5 → **wait**

15 Exotic options (SM 7.5 — 9 types)

Differ from plain vanilla in pay-off & time of exercise; more complex; mostly **OTC**.

Chooser	After a specified period buyer decides whether it is a call or a put
Compound	"Split-fee" / option on an option
Barrier	Activated only if underlying reaches a certain price in a set period
Asian	Pay-off on **average** price over a period
Look-back	Holder chooses most favourable strike from min / max price during life
Basket	Value depends on a portfolio (basket)
Bermuda	Compromise between European & American
Binary / digital	Pay-off on happening of a specific event
Spread	Pay-off on difference between prices of two underlyings

⑥ Index futures (SM 5.2)

- → Based on level of an index (Nifty 50, Sensex); used to **speculate** on the market or **hedge** a portfolio against decline
- → Not based on tangible goods → **all settlements in cash**; traded in number of contracts; value = index × multiplier (SM: 1 contract = 50 units × Nifty)
- → Margin is **continuous** — computed every business day; position squared off at closing price on expiry (SM: last Thursday of the month)
- → Purposes of futures trading (SM): **speculation** and **hedging**. Index futures hedge only **systematic** risk
- → SM TYK theory: reasons index futures became more popular than stock futures; short note on marking to market

⑯ Credit derivatives (SM 8)

- → **CDOs** — backed by pool of bonds, ABS, REITs, other CDOs (securitisation is backed by loans / receivables); covers CBOs & CLOs. **Types**: **Cash-flow CDO** (backed by cash-market debt; relies on collateral's cash to pay SPV securities) · **Synthetic CDO** (credit risk transferred **without** transferring assets; unfunded (CDS only) / fully funded (CLN) / partially funded) · **Arbitrage CDO** (issuer captures spread between collateral return and borrowing cost + management fee)
- → **Risks in CDOs**: default (hits equity / junior tranche first), interest-rate (basis), liquidity, prepayment, reinvestment, foreign-exchange
- → **CDS** = Credit (loan) + Default (non-payment) + Swap (exchange of risk): buyer pays periodic premium; seller pays on default. **Features**: non-standardised private (forward-type) contract; not exchange-traded; ISDA guidelines; can be bought from a third party; bondholder can protect against issuer insolvency; premium rises with risk. **Uses**: hedging, arbitrage, speculation. **Parties**: reference entity, buyer (investor), seller (writer). **Settlement**: physical (deliver bonds, get par) or cash (par − market value)

⑰ Commodity derivatives (SM 10)

- → **Necessary conditions (SM)**: commodity **durable & storable**; units **homogeneous**; **frequent price fluctuations with wide amplitude**; supply & demand **large**; supply flows naturally to market and there are **breakdowns in existing pattern of forward contracting**
- → **Investing**: complements investment in commodity-using companies; invest in a country's consumption & production; no dividends, only price returns. Market advantages: managers prefer derivatives to tangibles; less hassle (delivery); indirect real-asset exposure — inflation hedge
- → **Futures**: easiest & cheapest way; 3 categories — agricultural (soft), energy, metals. **Swaps**: **fixed-floating** and **commodity-for-interest**; valued as a strip of forwards; factors — hedging cost, institutional structure, liquidity, seasonality, futures bid / offer spread, brokerage, credit risk / capital / admin costs

18 Weather & electricity derivatives (SM 11–12)

→ **Weather**: underlying = weather measure (rainfall, temperature, humidity, wind); hedges **volume risk**, not price risk; first deal 1997 (Aquila Energy, OTC). **vs insurance**: insurance covers extreme low-probability events; weather derivatives cover normal-condition uncertainty too. Pricing issues: data reliability, weather forecasting, temperature modelling

→ **Electricity**: **forwards** (fixed quantity at forward price at maturity) · **futures** (standardised trading, settlement) · **swaps** (pay fixed vs floating price; = strip of forwards; locational basis swaps)

19 Derivative mishaps & 6 lessons (SM 13)

Orange County (1994)	Treasurer Robert Citron, no trading background; yield-curve play + ~$2.8 bn inverse floaters; leveraged ~$7.5 bn fund to ~$20 bn via repo; Fed raised rates 6 times in 1994 → loss **~$2 bn**; Citron jailed 1 yr, fined $1,00,000
Barings (1995)	Nick Leeson: arbitrage Nikkei 225 futures Singapore vs Osaka turned speculative long; also ran **back office**, hid losses in error account **'8888'**; loss $827 mn (SM also cites $1.4 bn); 233-year-old bank bankrupt; ING bought it for **£1**. Led to operational-risk capital in Basel II
P&G & Gibson Greetings (1994)	Bankers Trust sold complex **leveraged (ratio / power) swaps**; P&G lost $106 mn; Gibson $3 mn then $16.7 mn; out-of-court settlements; BT taken over by Deutsche Bank (1999)

Lessons: (a) don't buy a derivative you don't understand (value in-house) · (b) due diligence before making Treasury a profit centre (don't link treasurer's pay to profit) · (c) specify risk limits · (d) separate **front, middle & back** offices · (e) a hedger shouldn't become a speculator · (f) stress tests, scenario analysis (VaR + scenarios, 10–20 yrs data for black swans).

20 Correlation — how Ch 9 actually connects

Ch 9 idea	The actual link (app-written cross-reference; the Ch 9 side is SM)
Index-futures beta hedge	**Ch 6**: $\beta_p = \Sigma\, w_i\beta_i$ (weighted average of stock betas) goes straight into contracts = Value × $(\beta_2 - \beta_1)$ ÷ (futures price × lot). Hedge removes only **systematic** risk — unsystematic was already diversified away (Ch 6).
Cost of carry $F = S{\cdot}e^{rt}$	**Ch 10**: interest rate parity $F = S \times (1 + r_h) \div (1 + r_f)$ is the same idea — forward = spot + carry, else arbitrage. Actual F > fair F → buy spot, sell forward/futures (cash-and-carry), in both chapters.
Put-call parity: C + PV(K) = P + S	Used in ICAI answers to get P from C (May 2026: P = −S + C + PV(K)). **Ch 5**: warrant value (MP − E) × n is the **intrinsic value of a call**; a convertible = straight bond + call on shares.
Call / put payoffs	**Ch 10**: importer (payable) buys a **call** on the currency, exporter (receivable) buys a **put** — same max(S − K, 0) / max(K − S, 0) payoff, net of premium. **Ch 12**: interest-rate cap = series of calls on the rate (borrower), floor = series of puts (lender), collar = buy cap + sell floor.
Real options	**Ch 3**: plain NPV ignores managerial flexibility; option to expand / abandon / wait adds value → project worth NPV + option value. A negative-NPV project can still be accepted.
CDO tranches	**Ch 7**: same tranching as securitisation — senior, mezzanine, equity; losses hit the equity tranche first.
Mishaps & 6 lessons	**Ch 2**: risk limits, front/middle/back-office separation, VaR + stress tests = the risk-management framework. **FR (Ind AS 109)**: derivatives are at FVTPL unless designated in hedge accounting.

⑨ Derivatives Analysis and Valuation — what ICAI asked (Sep 2025, Jan 2026, May 2026)

45 marks set across the last three attempts (12.7% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.

Attempt	Question	Marks	Concept tested
Sep 2025	MCQ CS-III 7-9	6	Long strangle (buy 3-month call at 750 and put at 600 on the same stock): net gain/loss at expiry for unchanged, lower and higher price (MCQ)
Sep 2025	Q4(b)	4	Hedging a long and a short stock position with Nifty futures using betas: number of contracts and overall profit/loss on close-out (Numerical)
Jan 2026	Q1(c)	4	Long strangle (buy call at 620 and put at 480): profit/loss at three expiry prices and upper and lower break-even points (Numerical)
Jan 2026	Q2(c)	4	Real options versus financial options on four points: underlying asset, pay-off, exercise period, approach (Theory)
Jan 2026	Q3(a)	6	Long futures plus short call: profit/loss on futures and on option net of brokerage, overall profit and total brokerage (Numerical)
Jan 2026	Q6(b)	4	Real option: value of abandonment option by the risk-neutral method (up/down project values, abandonment price, risk-free rate) (Numerical)
May 2026	MCQ CS-II 4-6	6	Real option to wait: NPV today and after one year in high/low scenarios, risk-neutral returns and probability, PV of the option to wait (MCQ)
May 2026	Q2(b)	4	Six options (calls and puts, three strikes): ITM/ATM/OTM, intrinsic and time value, exercise or decline and net profit/loss per contract at +150 and -150 expiry prices (Numerical)
May 2026	Q6(b)	7	Black-Scholes: hedge ratio (N(d1)), probability of expiry above exercise price, value of call and put (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. What are the reasons for stock index futures becoming more popular financial derivatives over stock futures segment in India?

ICAI answer: "Please refer paragraph 5.2.4" – SM paragraph 5.2.4 reproduced

5.2.4 Advantages of Stock Index Futures Trading Vs. Stock Futures Trading

Stock index futures is most popular financial derivatives over stock futures due to following reasons:

1. It adds flexibility to one's investment portfolio. Institutional investors and other large equity holders prefer this instrument the most in terms of portfolio hedging purpose. The stock systems do not provide this flexibility and hedging.
2. It creates the possibility of speculative gains using leverage. Because a relatively small amount of margin money controls a large amount of capital represented in a stock index contract, a small change in the index level might produce a profitable return on one's investment if one is right about the direction of the market. Speculative gains in stock futures are limited but liabilities are greater.
3. Stock index futures are the most cost-efficient hedging device whereas hedging through individual stock futures is costlier.
4. Stock index futures cannot be easily manipulated whereas individual stock price can be exploited more easily.
5. Since, stock index futures consists of many securities, so being an average stock, is much less volatile than individual stock price. Further, it implies much lower capital adequacy and margin requirements in comparison of individual stock futures. Risk diversification is possible under stock index future than in stock futures.
6. One can sell contracts as readily as one buys them and the amount of margin required is the same.
7. In case of individual stocks the outstanding positions are settled normally against physical delivery of shares. In case of stock index futures they are settled in cash all over the world on the premise that index value is safely accepted as the settlement price.
8. It is also seen that regulatory complexity is much less in the case of stock index futures in comparison to stock futures.
9. It provides hedging or insurance protection for a stock portfolio in a falling market.

Theory Q2

2. Write short note on Marking to market.

ICAI answer: "Please refer paragraph 5.2.3" – SM paragraph 5.2.3 reproduced

5.2.3 Marking to Market

It implies the process of recording the investments in traded securities (shares, debt-instruments,

etc.) at a value, which reflects the market value of securities on the reporting date. In the context of derivatives trading, the futures contracts are marked to market on periodic (or daily) basis. Marking to market essentially means that at the end of a trading session, all outstanding contracts are repriced at the settlement price of that session. Unlike the forward contracts, the future contracts are repriced every day. Any loss or profit resulting from repricing would be debited or credited to the margin account of the broker. It, therefore, provides an opportunity to calculate the extent of liability on the basis of repricing. Thus, the futures contracts provide better risk management measure as compared to forward contracts.

Suppose on 1st day we take a long position, say at a price of ₹ 100 to be matured on 7th day. Now on 2nd day if the price goes up to ₹ 105, the contract will be repriced at ₹ 105 at the end of the trading session and profit of ₹ 5 will be credited to the account of the buyer. This profit of ₹ 5 may be drawn and thus cash flow also increases. This marking to market will result in three things – one, you will get a cash profit of ₹ 5; second, the existing contract at a price of ₹ 100 would stand cancelled; and third you will receive a new futures contract at ₹ 105. In essence, the marking to market feature implies that the value of the futures contract is set to zero at the end of each trading day.

Theory Q3

3. State any four assumptions of Black Scholes Model.

ICAI answer: "Please refer paragraph 7.3" – SM paragraph 7.3 reproduced

7.3 Black-Scholes Model

The Black-Scholes model is used to calculate a theoretical price of an Option. The Black-Scholes price is nothing more than the amount an option writer would require as compensation for writing a call and completely hedging the risk of buying stock. The important point is that the hedger's view about future stock prices is irrelevant. Thus, while any two investors may strongly disagree on the rate of return they expect on a stock they will, given agreement to the assumptions of volatility and the risk-free rate, always agree on the fair value of the option on that underlying asset. This key concept underlying the valuation of all derivatives -- that fact that the price of an option is independent of the risk preferences of investors -- is called risk-neutral valuation. It means that all derivatives can be valued by assuming that the return from their underlying assets is the risk-free rate.

The model is based on a normal distribution of underlying asset returns.

The following assumptions accompany the model:

1. European Options are considered,
2. No transaction costs,
3. Short term interest rates are known and are constant,
4. Stocks do not pay dividend,
5. Stock price movement is similar to a random walk,
6. Stock returns are normally distributed over a period of time, and
7. The variance of the return is constant over the life of an Option.

The original formula for calculating the theoretical option price (OP) is as follows:

Where:

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{v^2}{2}\right)t}{v\sqrt{t}}$$

$$d_2 = d_1 - v\sqrt{t}$$

The variables are:

S = current stock price

X = strike price of the option

t = time remaining until expiration, expressed as a percent of a year

r = current continuously compounded risk-free interest rate

v = annual volatility of stock price (the standard deviation of the short-term returns over one year).

ln = natural logarithm

N(x) = standard normal cumulative distribution function (Area under Normal Curve)

e = the exponential function

Understanding the formula

$N(d_1)$ represents the hedge ratio of shares of stock to Options necessary to maintain a fully hedged position.

Consider the Option holder as an investor who has borrowed an equivalent amount of the exercise price at interest rate r. $Xe^{-rt}N(d_2)$ represents this borrowing which is equivalent to the present value of the exercise price times an adjustment factor of $N(d_2)$

N(d2) in the above formulae represents the probability that price in spot market on expiration would be higher than the exercise price of the call option.

The main advantage of the Black-Scholes model is speed -- it lets you calculate a very large number of option prices in a very short time.

The Black-Scholes model has one major limitation that it cannot be used to accurately price options with an American-style exercise as it only calculates the option price at one point of time -- at expiration. It does not consider the steps along the way where there could be the possibility of early exercise of an American option.

Illustration 3

(i) The shares of TIC Ltd. are currently priced at ₹ 415 and call option exercisable in three months' time has an exercise rate of ₹ 400. Risk free interest rate is 5% p.a. and standard deviation (volatility) of share price is 22%. Based on the assumption that TIC Ltd. is not going to declare any dividend over the next three months, is the option worth buying for ₹ 25?

(ii) Calculate value of aforesaid call option based on Black Scholes valuation model if the current price is considered as ₹ 380.

(iii) What would be the worth of put option if current price is considered ₹ 380.

(iv) If TIC Ltd. share price at present is taken as ₹ 408 and a dividend of ₹ 10 is expected to be paid in the two months time, then, calculate value of the call option.

Given

$\ln(1.0375) = 0.03681$

$\ln(0.95) = -0.05129$

$\ln(0.9952) = -0.00481$

$e^{0.0125} = 1.012578$

$e^{0.008333} = 1.0084$

Solution

(i) Given: TIC Ltd. Current Price = ₹ 415

Exercise rate = 400

Risk free interest rate is = 5% p.a.

SD (Volatility) = 22%

Based on the above bit is calculated value of an option based on Black Scholes Model:

$$d_1 = \frac{\ln\left(\frac{415}{400}\right) + \left[.05 + \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}}$$

$$= \frac{.03681 + .01855}{.11} = .5032727$$

$$d_2 = \frac{\ln\left(\frac{415}{400}\right) + \left[.05 - \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}}$$

$$= \frac{.03681 + .00645}{.11} = 0.3932727$$

$N(d_1) = N(.50327) = 1 - 0.3072 = 0.6928$

$N(d_2) = N(.39327) = 1 - 0.3471 = 0.6529$

$$\text{Value of Option} = 415(.6928) - \frac{400}{e^{(.05)(.25)}}(.6529)$$

$$= 287.512 - \frac{400}{1.012578}(.6529) = 287.512 - 257.916 = ₹\ 29.60$$

NB : N(0.39327) can also be find as under :

Step 1: From table of area under normal curve find the area of variable 0.39 i.e. 0.6517.

Step 2: From table of area under normal curve find the area of variable 0.40.

Step 3: Find out the difference between above two variables and areas under normal curve.

Step 4 : Using interpolation method find out the value of 0.00327. Which is as follows:

$$\frac{0.0037}{0.01} \times 0.00327 = 0.0012$$

Step 5: Add this value, computed above to the N(0.39). Thus N (0.39327) = 0.6517 + 0.0012 = 0.6529

Since market price of ₹ 25 is less than ₹ 29.60 (as per Black Scholes Valuation model) indicate that option is underpriced, hence worth buying.

(ii) If the current price is taken as ₹ 380 the computations are as follows:

$$d_1 = \frac{\ln\left(\frac{380}{400}\right) + \left[.05 + \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}} = \frac{-0.05129 + .01855}{.11} = -0.297636$$

$$d_2 = \frac{\ln\left(\frac{380}{400}\right) + \left[.05 - \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}} = \frac{-0.05129 + .00645}{.11} = -0.407636$$

$$V_o = V_s\, N(d_1) - \frac{E}{e^{rt}}\, N(d_2)$$

$N(d_1) = N(-0.297636) = 0.3830$

$N(d_2) = N(-0.407636) = 0.3418$

$$380\ (.3830) - \frac{400}{e^{(.05)\,(.25)}} \times (.3418)$$

$$145.54 - \frac{400}{1.012578}\ (.3418) = 145.54 - 135.02 = ₹\ 10.52$$

(iii) Value of call option = ₹ 10.52

Current Market Value = ₹ 415

Present Value of Exercise Price = $\frac{400}{1.0125} = 395.06$ or $\frac{400}{1.012578} = 395.03$

Value of Put Option can be find by using Put Call Parity relationship as follows:

$V_p = -V_s + V_c + PV\ (E)$

$V_p = -380 + 10.52 + 395.06 = 25.58$

= ₹ 25.58 Ans

or -380 + 10.52 + 395.03 = 25.55

= ₹ 25.55

(iv) Since dividend is expected to be paid in two months time we have to adjust the share price and then use Black Scholes model to value the option:

Present Value of Dividend (using continuous discounting) = Dividend $\times\ e^{-rt}$

$= ₹\ 10 \times e^{-.05\ \times\ .16666}$

$= ₹\ 10 \times e^{-.008333}$

= ₹ 9.917 (Please refer Exponential Table)

Adjusted price of shares is ₹ 408 – 9.917 = ₹ 398.083

This can be used in Black Scholes model

$$d_1 = \frac{\ln\left(\frac{398.083}{400}\right) + \left[.05 + \frac{1}{2}\,(.22)^2\right].25}{.22\ \sqrt{.25}} = \frac{-.00481 + .01855}{.11} = 0.125$$

$$d_2 = \frac{\ln\left(\frac{398.083}{400}\right) + \left[.05 - \frac{1}{2}\,(.22)^2\right].25}{.22\ \sqrt{.25}} = \frac{-.00481 + .00645}{.11} = 0.015$$

$N(d_1) = N(0.125) = .5498$

$N(d_2) = N(0.015) = .5060$

$$\text{Value of Option} = 398.083\ \ (.5498) - \frac{400}{e_{(.05)\,(.25)}}\ (.5060)$$

$$218.866 - \frac{400}{e^{.0125}}\ (.5060)$$

$$218.866 - \frac{400}{1.012578}\ (.5060) = 218.866 - 199.8858 = ₹\ 18.98$$

Theory Q4

4. Define the term Greeks with respect to options.

ICAI answer: "Please refer paragraph 7.4" – SM paragraph 7.4 reproduced

7.4 Greeks

The Greeks are a collection of statistical values (expressed as percentages) that give the investor a better overall view of how a stock has been performing. These statistical values can be helpful in deciding what options strategies are best to use. The investor should remember that statistics show trends based on past performance. It is not guaranteed that the future performance of the stock will behave according to the historical numbers. These trends can change drastically based on new stock performance.

Before we discuss these statistical measures let us discuss the factors that affects the value of option as these statistical measures are related to changes in the in these factors.

7.4.1 Factors Affecting Value of an Option

There are a number of different mathematical formulae, or models, that are designed to compute the fair value of an option. You simply input all the variables (stock price, time, interest rates, dividends and future volatility), and you get an answer that tells you what an option should be worth. Here are the general effects the variables have on an option's price:

(a) ***Price Movement of the Underlying:*** The value of calls and puts are affected by changes in the underlying stock price in a relatively straightforward manner. When the stock price goes up, calls should gain in value and puts should decrease. Put options should increase in value and calls should drop as the stock price falls.

(b) ***Time till expiry:*** The option's future expiry, at which time it may become worthless, is an important and key factor of every option strategy. Ultimately, time can determine whether your option

trading decisions are profitable. To make money in options over the long term, you need to understand the impact of time on stock and option positions.

With stocks, time is a trader's ally as the stocks of quality companies tend to rise over long periods of time. But time is the enemy of the options buyer. If days pass without any significant change in the stock price, there is a decline in the value of the option. Also, the value of an option declines more rapidly as the option approaches the expiration day. That is good news for the option seller, who tries to benefit from time decay, especially during that final month when it occurs most rapidly.

(c) ***Volatility in Stock Prices:*** Volatility can be understood via a measure called Statistical (sometimes called historical) Volatility, or SV for short. SV is a statistical measure of the past price movements of the stock; it tells you how volatile the stock has actually been over a given period of time.

But to give you an accurate fair value for an option, option pricing models require you to put in what the future volatility of the stock will be during the life of the option. Naturally, option traders don't know what that will be, so they have to try to guess. To do this, they work the options pricing model "backwards" (to put it in simple terms). After all, you already know the price at which the option is trading; you can also find the other variables (stock price, interest rates, dividends, and the time left in the option) with just a bit of research. So, the only missing number is future volatility, which you can calculate from the equation.

(d) ***Interest Rate***- Another feature which affects the value of an Option is the time value of money. The greater the interest rates, the present value of the future exercise price are less.

Now let us discuss these measures.

7.4.2 Delta

A by-product of the Black-Scholes model is the calculation of the delta. It is the degree to which an option price will move given a small change in the underlying stock price. For example, option price (with a delta of 0.5) will move half a rupee for every full rupee movement in the underlying stock.

A deeply out-of-the-money call will have a delta very close to zero; a deeply in-the-money call will have a delta very close to 1.

The formula for a delta of a European call on a non-dividend paying stock is:

Delta = N (d_1) (see Black-Scholes formula above for d1)

Call Deltas are positive; Put Deltas are negative, reflecting the fact that the Put option price and the underlying stock price are inversely related. The Put Delta is equal to (Call Delta – 1).

As discussed earlier the delta is often called the Hedge Ratio. If you have a portfolio consisting short 'n' options (e.g., you have written n calls) and holding number of shares (units of the underlying i.e., n multiplied by the delta). This gives you a riskless position - i.e., a portfolio which would be worth the same whether the stock price rose by a very small amount or fell by a very small amount. In such a "delta neutral" portfolio any gain in the value of the shares held due to a rise in the share price would be exactly offset by a loss on the value of the calls written, and vice versa.

Note that as the Delta changes with the stock price and time to expiration the number of shares would need to be continually adjusted to maintain the hedge. How quickly the delta changes with the stock price are given by 'Gamma'.

In addition to Delta there are some other "Greeks" which some find useful when constructing option strategies.

7.4.3 Gamma

It measures how fast the Delta changes for small changes in the underlying stock price i.e. the Delta of the Delta. If you are hedging a portfolio using the Delta-hedge technique then you will want to keep gamma as small as possible, the smaller it is the less often you will have to adjust the hedge to maintain a delta neutral position. If gamma is too large, a small change in stock price could wreck your hedge. Adjusting gamma, however, can be tricky and is generally done using options i.e. it cannot be done by selling or buying underlying asset rather by selling or buying options.

7.4.4 Theta

It is change in the option price upon one day decrease in time to expiration. Basically, it is a measure of time decay. Unless you and your portfolio are travelling at close to the speed of light the passage of time is constant and inexorable. Thus, hedging a portfolio against time decay, the effects of which are completely predictable, would be pointless.

7.4.5 Rho

The change in option price given a one percentage point change in the risk-free interest rate. It is sensitivity of option value to change in interest rate. Rho indicates the absolute change in option value for a one percent change in the interest rate. For example, a Rho of 0.06 indicates the option's theoretical value will increase by 0.06 if the interest rate is decreased by 1.0.

7.4.6 Vega

Sensitivity of option value to change in volatility. Vega indicates an absolute change in option value for a one percent change in volatility. For example, a Vega of 0.09 indicates an absolute change in the option's theoretical value will increase by 0.09 if the volatility percentage is increased by 1.0 or decreased by 0.09 if the volatility percentage is decreased by 1.0. Results may not be exact due to rounding. It can also be stated as the change in option price given a one percentage point change in volatility. Like delta and gamma, Vega is also used for hedging.

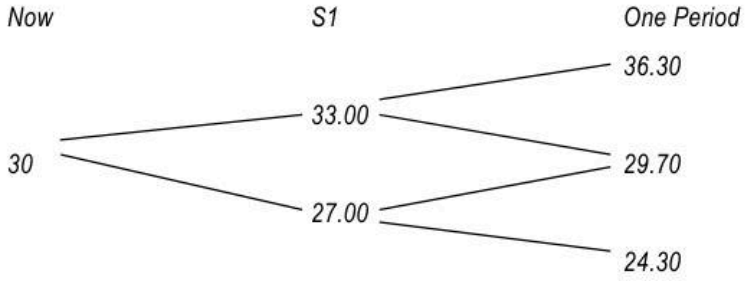
Ⓘ SM illustrations with ICAI solutions

7. Option Valuation Techniques

SM Ch 9 · Illustration 1 (under 7.1 Binomial Model)

Illustration 1

Following is a two sub-periods tree of 6-months each for the share of CAB Ltd.:



Using the binomial model, calculate the current fair value of a regular call option on CAB Stock with the following characteristics: X = ₹ 28, Risk Free Rate = 5 percent p.a. You should also indicate the composition of the implied riskless hedge portfolio at the valuation date.

Solution — ICAI

Solution

u = 33.00/30.00 = 36.30/33.00 = 1.10 d = 27.00/30.00 = 24.30/27.00 = 0.90

$r = (1 + .05)^{1/2} = 1.0247$

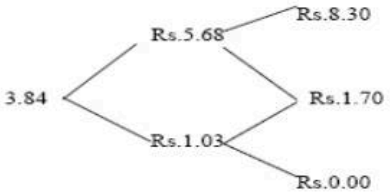
$$p = \frac{r-d}{u-d} = \frac{1.0247-0.90}{1.10-0.90} = 0.1247/0.20 = 0.6235 \text{ (Prob. of increase in Price of Share)}$$

Prob. of decrease in Price of Share = 1 – 0.6235 = 0.3765

C_{uu} = Max [0, 36.30 – 28] = 8.30

C_{ud} = Max [0, 29.70 – 28] = 1.70

C_{dd} = Max [0, 24.30 – 28] = 0



$$C_U = \frac{(0.6235)(8.30)+(0.3765)(1.70)}{1.025} = \frac{5.175+0.640}{1.025} = 5.815/1.025 = ₹\ 5.673$$

$$C_d = \frac{(0.6235)(1.70)+(0.3765)(0.00)}{1.025} = \frac{1.05995}{1.025} = ₹\ 1.0341$$

$$C_o = \frac{(0.6235)(5.673)+(0.3765)(1.0341)}{1.025} = \frac{3.537+0.3893}{1.025} = ₹\ 3.83$$

The composition of the implied risk-less hedge portfolio at valuation date is called Delta (Δ) and it is calculated as follows:

$$\frac{C_u - C_d}{S_U - S_d}$$

Where,

C_u = Pay-off from Call Option if price of Stock goes up

C_d = Pay-off from Call Option if price of Stock goes down

S_u = Upward price of the Stock

S_d = Downward price of the Stock

Accordingly, the Risk-less Portfolio shall require Δ Share shall be required for writing off one Call Option. The Δ shall be computed as follows:

$$\Delta = \frac{5-0}{33-27} = \frac{5}{6}$$

Thus, $\frac{5}{6}$ shares shall be held or purchased for writing one Call Option.

SM Ch 9 · Illustration 2 (under 7.2 Risk Neutral Method)

Illustration 2

The current market price of an equity share of Penchant Ltd is ₹ 420. Within a period of 3 months, the maximum and minimum price of it is expected to be ₹ 500 and ₹ 400 respectively. If the risk free rate of interest be 8% p.a., what should be the value of a 3 months Call option under the "Risk Neutral" method at the strike rate of ₹ 450 ? Given $e^{0.02}$ = 1.0202

Solution — ICAI

Solution

Let the probability of attaining the maximum price be p

(500 - 420) x p+(400 - 420) x (1-p) = 420 x ($e^{0.02}$.1)

or, 80p - 20(1 - p) = 420 x 0.0202

or, 80p – 20 + 20p = 8.48

or, 100p = 28.48

p= 0.2848

The value of Call Option in ₹ = $\frac{0.2848x(500-450)}{1.0202} = \frac{0.2848x50}{1.0202}$ =13.96

SM Ch 9 · Illustration 3 (under 7.3 Black-Scholes Model)

Illustration 3

(i) The shares of TIC Ltd. are currently priced at ₹ 415 and call option exercisable in three months' time has an exercise rate of ₹ 400. Risk free interest rate is 5% p.a. and standard deviation (volatility) of share price is 22%. Based on the assumption that TIC Ltd. is not going to declare any dividend over the next three months, is the option worth buying for ₹ 25?

(ii) Calculate value of aforesaid call option based on Black Scholes valuation model if the current price is considered as ₹ 380.

(iii) What would be the worth of put option if current price is considered ₹ 380.

(iv) If TIC Ltd. share price at present is taken as ₹ 408 and a dividend of ₹ 10 is expected to be paid in the two months time, then, calculate value of the call option.

Given

ln(1.0375) = 0.03681

ln(0.95) = -0.05129

ln(0.9952) = -0.00481

$e^{0.0125}$ = 1.012578

$e^{0.008333}$ = 1.0084

Solution — ICAI

Solution

(i) Given: TIC Ltd. Current Price = ₹ 415

Exercise rate = 400

Risk free interest rate is = 5% p.a.

SD (Volatility) = 22%

Based on the above bit is calculated value of an option based on Black Scholes Model:

$$d_1 = \frac{\ln\left(\frac{415}{400}\right) + \left[.05 + \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}}$$

$$= \frac{.03681 + .01855}{.11} = .5032727$$

$$d_2 = \frac{\ln\left(\frac{415}{400}\right) + \left[.05 - \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}}$$

$$= \frac{.03681 + .00645}{.11} = 0.3932727$$

$N(d_1) = N(.50327) = 1 - 0.3072 = 0.6928$

$N(d_2) = N(.39327) = 1 - 0.3471 = 0.6529$

$$\text{Value of Option} = 415(.6928) - \frac{400}{e^{(.05)(.25)}}(.6529)$$

$$= 287.512 - \frac{400}{1.012578}(.6529) = 287.512 - 257.916 = ₹\ 29.60$$

NB : N(0.39327) can also be find as under :

Step 1: From table of area under normal curve find the area of variable 0.39 i.e. 0.6517.

Step 2: From table of area under normal curve find the area of variable 0.40.

Step 3: Find out the difference between above two variables and areas under normal curve.

Step 4 : Using interpolation method find out the value of 0.00327. Which is as follows:

$$\frac{0.0037}{0.01} \times 0.00327 = 0.0012$$

Step 5: Add this value, computed above to the N(0.39). Thus N (0.39327) = 0.6517 + 0.0012 = 0.6529

Since market price of ₹ 25 is less than ₹ 29.60 (as per Black Scholes Valuation model) indicate that option is underpriced, hence worth buying.

(ii) If the current price is taken as ₹ 380 the computations are as follows:

$$d_1 = \frac{\ln\left(\frac{380}{400}\right) + \left[.05 + \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}} = \frac{-0.05129 + .01855}{.11} = -0.297636$$

$$d_2 = \frac{\ln\left(\frac{380}{400}\right) + \left[.05 - \frac{1}{2}(.22)^2\right].25}{.22\sqrt{.25}} = \frac{-0.05129 + .00645}{.11} = -0.407636$$

$$V_o = V_s\, N(d_1) - \frac{E}{e^{rt}} N(d_2)$$

$N(d_1) = N(-0.297636) = 0.3830$

$N(d_2) = N(-0.407636) = 0.3418$

$$380\ (.3830) - \frac{400}{e^{(.05)\,(.25)}} \times (.3418)$$

$$145.54 - \frac{400}{1.012578}\ (.3418) = 145.54 - 135.02 = ₹\ 10.52$$

(iii) Value of call option = ₹ 10.52

Current Market Value = ₹ 415

Present Value of Exercise Price = $\frac{400}{1.0125} = 395.06$ or $\frac{400}{1.012578} = 395.03$

Value of Put Option can be find by using Put Call Parity relationship as follows:

$V_p = -V_s + V_c + PV\ (E)$

$V_p = -380 + 10.52 + 395.06 = 25.58$

= ₹ 25.58 Ans

or -380 + 10.52 + 395.03 = 25.55

= ₹ 25.55

(iv) Since dividend is expected to be paid in two months time we have to adjust the share price and then use Black Scholes model to value the option:

Present Value of Dividend (using continuous discounting) = Dividend $\times\ e^{-rt}$

$= ₹\ 10 \times e^{-.05\ \times\ .16666}$

$= ₹\ 10 \times e^{-.008333}$

= ₹ 9.917 (Please refer Exponential Table)

Adjusted price of shares is ₹ 408 – 9.917 = ₹ 398.083

This can be used in Black Scholes model

$$d_1 = \frac{l_n\left(\frac{398.083}{400}\right) + \left[.05 + \frac{1}{2}(.22)^2\right].25}{.22\ \sqrt{.25}} = \frac{-.00481 + .01855}{.11} = 0.125$$

$$d_2 = \frac{l_n\left(\frac{398.083}{400}\right) + \left[.05 - \frac{1}{2}(.22)^2\right].25}{.22\ \sqrt{.25}} = \frac{-.00481 + .00645}{.11} = 0.015$$

$N(d_1) = N(0.125) = .5498$

$N(d_2) = N(0.015) = .5060$

$$\text{Value of Option} = 398.083\ (.5498) - \frac{400}{e_{(.05)\,(.25)}}\ (.5060)$$

$$218.866 - \frac{400}{e^{.0125}}\ (.5060)$$

$$218.866 - \frac{400}{1.012578}\ (.5060) = 218.866 - 199.8858 = ₹\ 18.98$$

9. Real Options

SM Ch 9 · Illustration 4 (under 9.3.1 Growth Options)

Illustration 4

ABC Ltd. is a pharmaceutical company possessing a patent of a drug called 'Aidrex', a medicine for aids patient. Being an approach drug ABC Ltd. holds the right of production of drugs and its marketing. The period of patent is 15 years after which any other pharmaceutical company produce the drug with same formula. It is estimated that company shall require to incur $ 12.5 million for development and market of the drug. As per a survey conducted the expected present value of cashflows from the sale of drug during the period of 15 years shall be $ 16.7 million. Cash flow from the previous similar type of drug have exhibited a variance of 26.8% of the present value of cashflows. The current yield on Treasury Bonds of similar duration (15 years) is 7.8%.

Determine the value of the patent.

Given

ln(1.336) = 0.2897

$e^{-1.0005} = 0.3677$ and $e^{-1.17} = 0.3104$

Solution — ICAI

Solution

The given solution is like valuation of stock option wherein delay in introduction of drug 'Aidrex' shall cause the loss of cashflow which is like payment of dividend.

To value the patent, we shall use Black Scholes Model for option pricing as follows:

Inputs

S (Spot Price)	= The Present Value of Cashflows = $16.7 million
E (Exercise Price)	= Cost of Development Formula = $ 12.5 million
σ^2(Variance of Cash flow)	= 26.8% i.e. 0.268
R (Risk Free Rate of Return)	= 7.8%

D (Expected cost of Delays) $= \frac{1}{15} = 0.0667$ i.e. 6.67%

Value call option

$C = S\,N(d_1)e^{-df} - Ee^{-rt}\,N(d_2)$

$$d_1 = \frac{\ln(S/E) + [R - D + (\frac{1}{2})\sigma^2]\ t}{\sigma\sqrt{t}}$$

$d_2 = d_1 - \sigma\sqrt{t}$

Accordingly,

$$d_1 = \frac{\ln(16.7/12.5) + [0.078 - 0.0667 + (\frac{1}{2})(0.268)]\ 15}{\sqrt{0.268}\ \sqrt{15}}$$

$$d_1 = \frac{0.2897 + (0.1453)15}{0.5177 \times 3.8730}$$

$$d_1 = \frac{0.2897 + 2.1795}{2.005}$$

$$d_1 = \frac{2.4692}{2.005} = 1.2315$$

$d_2 = 1.2315 - 2.005$

$d_2 = -0.7735$

$N(d_1) = 0.8910$

$N(d_2) = 0.2196$

Value of Patent

$= 16.7 \times e^{-0.0667 \times 15} \times 0.8910 - 12.5 \times e^{-0.078 \times 15} \times 0.2196$

$= 16.7 \times 0.3677 \times 0.8910 - 12.5 \times 0.3104 \times 0.2196$

$= 5.471 - 0.852 = 4.619$

Thus, the value of patents is $ 4.619 million

SM Ch 9 · Illustration 5 (under 9.3.2 Abandonment Option)

Illustration 5

IPL already in production of Fertilizer is considering a proposal of building a new plant to produce pesticides. Suppose the PV of proposal is ₹ 100 crore without the abandonment option. However, if market conditions for pesticide turns out to be favourable the PV of proposal shall increase by 30%. On the other hand, market conditions remain sluggish the PV of the proposal shall be reduced by 40%. In case company is not interested in continuation of the project it can be disposed of for ₹ 80 crore.

If the risk-free rate of interest is 8% then what will be value of abandonment option.

Solution — ICAI

Solution

Decision Tree showing pay off

Year 0	Year 1	Option Pay off
100	130	0
	60	80-60 = 20

First of all we shall calculate probability of high demand (p) using risk neutral method as follows:

8% = p x 30% + (1-p) x (-40%)

0.08 = 0.30 p - 0.40 + 0.40p

$$p = \frac{0.48}{0.70} = 0.686$$

The value of abandonment option will be as follows:

Expected Payoff at Year 1

= p x 0 + [(1-p) x 20]

= 0.686 x 0 + [0.314 x 20] = ₹ 6.28 crore

Since expected pay off at year 1 is ₹ 6.28 crore. Present value of expected pay off will be:

$$\frac{6.28}{1.08} = ₹ 5.81 \text{ crore.}$$

This is the value of abandonment option (Put Option).

SM Ch 9 · Illustration 6 (under 9.3.3 Timing Option)

Illustration 6

Suppose MIS Ltd. is considering installation of solar electricity generating plant for light the staff quarters. The plant shall cost ₹ 2.50 crore and shall lead to saving in electricity expenses at the current tariff by ₹ 21 lakh per year forever.

However, after one year with change in Government in state, the rate of electricity is subject to change. Accordingly, the saving in electricity can be of ₹ 12 lakh or ₹ 35 lakh per year and forever.

Assuming WACC of MIS Ltd. is 10% and risk-free rate of rate of return is 8%.

Decide whether MIS Ltd. should accept the project or wait and see.

Solution — ICAI

Solution

Here we shall evaluate NPV in two possible situations:

(1) As on Today

At cost of Capital of 10%, the value of saving forever = $\frac{₹\ 21\ lakhs}{0.10}$ = ₹ 2.1 crore

NPV = ₹ 2.1 crore - ₹ 2.5 crore = - ₹ 0.4 crore

Since NPV is negative, it does not worth to accept the project.

(2) After one Year

After one year these are two possible situations, either rate of electricity decreases or increase.

(a) If price of electricity increases

At cost of Capital of 10%, the value of saving forever = $\frac{₹35\ lakh}{0.10}$ = ₹ 3.50 crore

The position of the NPV will be as follows:

= ₹ 3.50 crore - ₹ 2.50 crore = ₹ 1 crore

And Rate of Return will be (3.5/2.5) - 1 = 0.40 is 40%

(b) If the price of electricity decreases, then value of saving forever will be

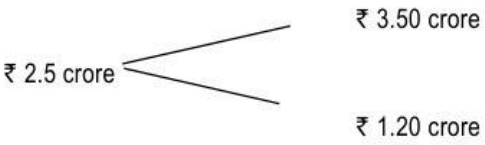
At cost of Capital of 10%, the value of saving forever = $\frac{12\ lakh}{0.10}$ = ₹ 1.20 crore

The position of the NPV will be as follows:

= ₹ 1.20 crore - ₹ 2.5 crore = - ₹ 1.3 crore

and Rate of Return will be (1.2/2.5) - 1 = -0.52 i.e. – 52.00%

Diagrammatically it can be shown below:



Let prob. of price increase be p. Then using Risk Neutral Method, the risk-free rate of return will be equal to expected saving as follows:

p x 0.40 + (1-p) (-0.52) = 0.08

0.40p - 0.52 + 0.52p = 0.08

0.92p = 0.60

p = 0.652

Hence, expected pay off = 0.652 x ₹ 1 crore + 0.348 x (- ₹ 1.30 crore) = ₹ 19.96 lakh.

PV of Pay off after one year = ₹ 19.96 lakh/ 1.08 = ₹ 18.48 lakh or ₹ 19.96 lakh/ 1.10 = ₹ 18.15 lakh.

Thus, it shall be advisable to wait and see as NPV may turn out to be positive after one year.

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. The 6-months forward price of a security is ₹ 208.18. The borrowing rate is 8% per annum payable with monthly rests. What should be the spot price?

Solution — ICAI

1. Calculation of spot price

The formula for calculating forward price is:

$$A = P\left(1+\frac{r}{n}\right)^{nt}$$

Where A = Forward price

P = Spot Price
r = rate of interest
n = no. of compounding
t = time

Using the above formula,

$208.18 = P\ (1 + 0.08/12)^{6}$

Or $208.18 = P \times 1.0409$

$P = 208.18/1.0409 = 200$

Hence, the spot price should be ₹ 200.

Practical Q2

2. The following data relate to Anand Ltd.'s share price:

Current price per share	₹ 1,800
6 months future's price/share	₹ 1,950

Assuming it is possible to borrow money in the market for transactions in securities at 12% per annum, you are required:

(i) to calculate the theoretical minimum price of a 6-months forward purchase; and

(ii) to explain arbitrage opportunity.

Solution — ICAI

2. Anand Ltd

(i) Calculation of theoretical minimum price of a 6 months forward contract-

Theoretical minimum price = ₹ 1,800 + (₹ 1,800 x 12/100 x 6/12) = ₹ 1,908

(ii) Arbitrage Opportunity-

The arbitrageur can borrow money @ 12 % for 6 months and buy the shares at ₹ 1,800. At the same time he can sell the shares in the futures market at ₹ 1,950. On the expiry date 6 months later, he could deliver the share and collect ₹ 1,950 pay off ₹ 1,908 and record a profit of ₹ 42 (₹ 1,950 – ₹ 1,908)

Practical Q3

3. On 31-8-2011, the value of stock index was ₹ 2,200. The risk-free rate of return has been 8% per annum. The dividend yield on this Stock Index is as under:

Month	Dividend Paid p.a.
January	3%
February	4%
March	3%
April	3%
May	4%
June	3%
July	3%
August	4%
September	3%
October	3%
November	4%
December	3%

Assuming that interest is continuously compounded daily, find out the future price of contract deliverable on 31-12-2011. Given: $e^{0.01583} = 1.01593$

Solution — ICAI

3. The duration of future contract is 4 months. The average yield during this period will be:

$$\frac{3\% + 3\% + 4\% + 3\%}{4} = 3.25\%$$

As per Cost to Carry model the future price will be

$F = Se^{(r_f - D)t}$

Where S = Spot Price

r_f = Risk Free interest

D = Dividend Yield

t = Time Period

Accordingly, future price will be

$= ₹ 2,200\ e^{(0.08 - 0.0325) \times 4/12} = ₹ 2,200\ e^{0.01583}$

= ₹ 2,200 x 1.01593 = ₹ 2235.05

Practical Q4

4. Calculate the price of 3 months PQR futures, if PQR (FV ₹10) quotes ₹220 on NSE and the three months future price quotes at ₹230 and the one month borrowing rate is given as 15 percent per annum and the expected annual dividend is 25 percent, payable before expiry. Also examine arbitrage opportunities.

Solution — ICAI

4. Future's Price = Spot + cost of carry – Dividend

$$F = 220 + 220 \times (0.15 \times 0.25)^{\dagger} - 0.25^{**} \times 10 = 225.75$$

** Entire 25% dividend is payable before expiry, which is ₹2.50.

† Alternatively monthly compounding can also be used.

Thus, we see that futures price by calculation is ₹225.75 which is quoted at ₹230 in the exchange.

(i) Analysis:

Fair value of Futures less than Actual futures Price:

Futures Overvalued Hence it is advised to sell. Also do Arbitraging by buying stock in the cash market.

Step I

He will buy PQR Stock at ₹220 by borrowing at 15% for 3 months. Therefore, his outflows are:

Cost of Stock	220.00
Add: Interest @ 15 % for 3 months i.e. 0.25 years (220 × 0.15 × 0.25)	8.25
Total Outflows (A)	228.25

Step II

He will sell March 2000 futures at ₹230. Meanwhile he would receive dividend for his stock.

Hence his inflows are	230.00
Sale proceeds of March 2000 futures	2.50
Total inflows (B)	232.50

Inflow – Outflow = Profit earned by Arbitrageur

= 232.50 – 228.25 = 4.25

Practical QS

5.

BSE	5000
Value of portfolio	₹ 10,10,000
Risk free interest rate	9% p.a.
Dividend yield on Index	6% p.a.
Beta of portfolio	1.5

We assume that a future contract on the BSE index with four months maturity is used to hedge the value of portfolio over next three months. One future contract is for delivery of 50 times the index.

Based on the above information calculate:

(i) Price of future contract.

(ii) The gain on short futures position if index turns out to be 4,500 in three months.

Solution — ICAI

5. **(i)** Current future price of the index = $5000 + 5000\ (0.09-0.06)\frac{4}{12} = 5000 + 50 = 5,050$

$\therefore$ Price of the future contract = ₹ 50 x 5,050 = ₹ 2,52,500

(ii) Hedge ratio = $\frac{1010000}{252500} \times 1.5 = 6$ contracts

Index after there months turns out to be 4500

Future price will be = $4500 + 4500\ (0.09-0.06) \times \frac{1}{12} = 4,511.25$

Therefore, Gain from the short futures position is = 6 x (5050 – 4511.25) x 50

= ₹1,61,625

Note: Alternatively we can also use daily compounding (exponential) formula.

Practical Q6

6. The share of X Ltd. is currently selling for ₹ 300. Risk free interest rate is 0.8% per month. A three-month futures contract is selling for ₹ 312. Develop an arbitrage strategy and show what your riskless profit will be 3 months hence assuming that X Ltd. will not pay any dividend in the next three months.

Solution — ICAI

6. The appropriate value of the 3 months futures contract is –

Fo = ₹ 300 $(1.008)^3$ = ₹ 307.26

Since the futures price exceeds its appropriate value it pays to do the following:-

Action	Initial Cash flow	Cash flow at time T (3 months)
Borrow ₹ 300 now and repay with interest after 3 months	+ ₹ 300	- ₹ 300 $(1.008)^3$ =- ₹ 307.26
Buy a share	- ₹ 300	ST
Sell a futures contract (Fo = 312/-)	0	₹ 312 – ST
Total	₹ 0	₹ 4.74

Such an action would produce a risk less profit of ₹ 4.74.

Practical Q7

7. A Mutual Fund is holding the following assets in ₹ Crores :

Investments in diversified equity shares	90.00
Cash and Bank Balances	10.00
	100.00

The Beta of the equity shares portfolio is 1.1. The index future is selling at 4300 level. The Fund Manager apprehends that the index will fall at the most by 10%. How many index futures he should short for perfect hedging? One index future consists of 50 units.

Substantiate your answer assuming the Fund Manager's apprehension will materialize.

Solution — ICAI

7. Number of index future to be sold by the Fund Manager is:

$$\frac{1.1\times 90,00,00,000}{4,300\times 50} = 4,605$$

Justification of the answer:

Loss in the value of the portfolio if the index falls by 10% is ₹ $\frac{11}{100}$ x90 Crore = ₹ 9.90 Crore.

Gain by short covering of index future is: $\frac{0.1\times 4,300\times 50\times 4,605}{1,00,00,000}$ = 9.90 Crore

This justifies the answer. Further, cash is not a part of the portfolio.

Practical Q8

8. A trader is having in its portfolio shares worth ₹ 85 lakhs at current price and cash ₹ 15 lakhs. The beta of share portfolio is 1.6. After 3 months the price of shares dropped by 3.2%.

Determine:

(i) Current portfolio beta

(ii) Portfolio beta after 3 months if the trader on current date goes for long position on ₹ 100 lakhs Nifty futures.

Solution — ICAI

8. (i) Current Portfolio Beta

Current Beta for share portfolio = 1.6

Beta for cash = 0

Current portfolio beta = 0.85 x 1.6 + 0 x 0.15 = 1.36

(ii) Portfolio beta after 3 months:

$$\text{Beta for portfolio of shares} = \frac{\text{Change in value of portfolio of share}}{\text{Change in value of market portfolio (Index)}}$$

$$1.6 = \frac{0.032}{\text{Change in value of market portfolio (Index)}}$$

Change in value of market portfolio (Index) = (0.032 / 1.6) x 100 = 2%

Position taken on 100 lakh Nifty futures : Long

Value of index after 3 months = ₹ 100 lakh x (1.00 - 0.02)

= ₹ 98 lakh

Mark-to-market paid = ₹ 2 lakh

Cash balance after payment of mark-to-market = ₹ 13 lakh

Value of portfolio after 3 months = ₹85 lakh x (1 - 0.032) + ₹13 lakh

= ₹95.28 lakh

$$\text{Change in value of portfolio} = \frac{₹100 \text{ lakh} - ₹95.28 \text{ lakh}}{₹100 \text{ lakh}} = 4.72\%$$

Portfolio beta = 0.0472/0.02 = 2.36

Practical Q9

9. Which position on the index future gives a speculator, a complete hedge against the following transactions:

(i) The share of Right Limited is going to rise. He has a long position on the cash market of ₹ 50 lakhs on the Right Limited. The beta of the Right Limited is 1.25.

(ii) The share of Wrong Limited is going to depreciate. He has a short position on the cash market of ₹ 25 lakhs on the Wrong Limited. The beta of the Wrong Limited is 0.90.

(iii) The share of Fair Limited is going to stagnant. He has a short position on the cash market of ₹ 20 lakhs of the Fair Limited. The beta of the Fair Limited is 0.75.

Solution — ICAI

9.

Sl. No. (1)	Company Name (2)	Trend (3)	Amount (₹) (4)	Beta (5)	(₹) (6) [(4) x (5)]	Position (7)
(i)	Right Ltd.	Rise	50 lakh	1.25	62,50,000	Short
(ii)	Wrong Ltd.	Depreciate	25 lakh	0.90	22,50,000	Long
(iii)	Fair Ltd.	Stagnant	20 lakh	0.75	15,00,000	Long
					25,00,000	Short

Practical Q10

10. Ram buys 10,000 shares of X Ltd. at a price of ₹ 22 per share whose beta value is 1.5 and sells 5,000 shares of A Ltd. at a price of ₹ 40 per share having a beta value of 2. He obtains a complete hedge by Nifty futures at ₹ 1,000 each. He closes out his position at the closing price of the next day when the share of X Ltd. dropped by 2%, share of A Ltd. appreciated by 3% and Nifty futures dropped by 1.5%.

What is the overall profit/loss to Ram?

Solution — ICAI

10. No. of the Future Contract to be obtained to get a complete hedge

$$= \frac{10000 \times ₹22 \times 1.5 - 5000 \times ₹40 \times 2}{₹1000}$$

$$= \frac{₹3,30,000 - ₹4,00,000}{₹1000} = 70 \text{ contracts}$$

Thus, by purchasing 70 Nifty future contracts to be long to obtain a complete hedge.

Cash Outlay

= 10000 x ₹ 22 – 5000 x ₹ 40 + 70 x ₹ 1,000

= ₹ 2,20,000 – ₹ 2,00,000 + ₹ 70,000 = ₹ 90,000

Cash Inflow at Close Out

= 10000 x ₹ 22 x 0.98 – 5000 x ₹ 40 x 1.03 + 70 x ₹ 1,000 x 0.985

= ₹ 2,15,600 – ₹ 2,06,000 + ₹ 68,950 = ₹ 78,550

Gain/ Loss

= ₹ 78,550 – ₹ 90,000 = - ₹ 11,450 (Loss)

Practical QII

11. On January 1, 2013 an investor has a portfolio of 5 shares as given below:

Security	Price	No. of Shares	Beta
A	349.30	5,000	1.15
B	480.50	7,000	0.40
C	593.52	8,000	0.90
D	734.70	10,000	0.95
E	824.85	2,000	0.85

The cost of capital to the investor is 10.5% per annum.

You are required to calculate:

(i) The beta of his portfolio.

(ii) The theoretical value of the NIFTY futures for February 2013.

(iii) The number of contracts of NIFTY the investor needs to sell to get a full hedge until February for his portfolio if the current value of NIFTY is 5900 and NIFTY futures have a minimum trade lot requirement of 200 units. Assume that the futures are trading at their fair value.

(iv) The number of future contracts the investor should trade if he desires to reduce the beta of his portfolios to 0.6.

No. of days in a year be treated as 365.

Given: ln (1.105) = 0.0998 and $e^{(0.015858)}$ = 1.01598

Solution — ICAI

11. (i) **Calculation of Portfolio Beta**

Security	Price of the Stock	No. of shares	Value	Weightage w_i	Beta B_i	Weighted Beta
A	349.30	5,000	17,46,500	0.093	1.15	0.107
B	480.50	7,000	33,63,500	0.178	0.40	0.071
C	593.52	8,000	47,48,160	0.252	0.90	0.227
D	734.70	10,000	73,47,000	0.390	0.95	0.370
E	824.85	2,000	16,49,700	0.087	0.85	0.074
			1,88,54,860			0.849

Portfolio Beta = 0.849

(ii) Calculation of Theoretical Value of Future Contract

Cost of Capital = 10.5% p.a. Accordingly, the Continuously Compounded Rate of Interest ln (1.105) = 0.0998

For February 2013 contract, t= 58/365= 0.1589

Further $F = Se^{rt}$

$F = ₹\ 5,900e^{(0.0998)(0.1589)}$

$F = ₹\ 5,900e^{0.015858}$

F= ₹ 5,900X1.01598 = ₹ 5,994.28

Alternatively, it can also be taken as follows:

$= ₹\ 5900\ e^{0.105\times58/365}$

$= ₹\ 5900\ e^{0.01668}$

$= ₹\ 5900 \times 1.01682 = ₹\ 5,999.24$

(iii) When total portfolio is to be hedged:

$$= \frac{\text{Value of Spot Position requiring hedging}}{\text{Value of Future Contract}} \times \text{Portfolio Beta}$$

$$= \frac{1,88,54,860}{5994.28 \times 200} \times 0.849 = 13.35 \text{ contracts say 13 or 14 contracts}$$

(iv) When total portfolio beta is to be reduced to 0.6:

$$\text{Number of Contracts to be sold} = \frac{P(\beta_P - \beta_P')}{F}$$

$$= \frac{1,88,54,860\,(0.849 - 0.600)}{5994.28 \times 200} = 3.92 \text{ contracts say 4 contracts}$$

Practical Q12

12. Details about portfolio of shares of an investor is as below:

Shares	No. of shares (lakh)	Price per share	Beta
A Ltd.	3.00	₹ 500	1.40
B Ltd.	4.00	₹ 750	1.20
C Ltd.	2.00	₹ 250	1.60

The investor thinks that the risk of portfolio is very high and wants to reduce the portfolio beta to 0.91. He is considering two below mentioned alternative strategies:

(i) Dispose off a part of his existing portfolio to acquire risk free securities, or

(ii) Take appropriate position on Nifty Futures which are currently traded at 8125 and each Nifty points is worth ₹ 200.

You are required to determine:

(1) portfolio beta,

(2) the value of risk free securities to be acquired,

(3) the number of shares of each company to be disposed off,

(4) the number of Nifty contracts to be bought/sold; and

(5) the value of portfolio beta for 2% rise in Nifty.

Solution — ICAI

12.

Shares	No. of shares (lakhs) (1)	Market Price of Per Share (2)	× (2) (₹ lakhs)	% to total (w)	ß (x)	wx
A Ltd.	3.00	500.00	1500.00	0.30	1.40	0.42
B Ltd.	4.00	750.00	3000.00	0.60	1.20	0.72
C Ltd.	2.00	250.00	500.00	0.10	1.60	0.16
			5000.00	1.00		1.30

(1) Portfolio beta 1.30

(2) Required Beta 0.91

Let the proportion of risk free securities for target beta 0.91 = p

$0.91 = 0 \times p + 1.30 (1 - p)$

$p = 0.30$ i.e. 30%

Shares to be disposed off to reduce beta (5000 × 30%) ₹ 1,500 lakh and Risk Free securities to be acquired.

(3) Number of shares of each company to be disposed off

Shares	% to total (w)	Proportionate Amount (₹ lakhs)	Market Price Per Share	No. of Shares (Lakh)
A Ltd.	0.30	450.00	500.00	0.90
B Ltd.	0.60	900.00	750.00	1.20
C Ltd.	0.10	150.00	250.00	0.60

(4) Number of Nifty Contract to be sold

$$\frac{(1.30-0.91) \times 5000 \text{ lakh}}{8{,}125 \times 200} = 120 \text{ contracts}$$

(5) 2% rises in Nifty is accompanied by 2% x 1.30 i.e. 2.6% rise for portfolio of shares

	₹ Lakh
Current Value of Portfolio of Shares	5000
Value of Portfolio after rise	5130
Mark-to-Market Margin paid (8125 × 0.020 × ₹ 200 × 120)	39
Value of the portfolio after rise of Nifty	5091
% change in value of portfolio (5091 – 5000)/ 5000	1.82%
% rise in the value of Nifty	2%
Beta	0.91

Practical Q13

13. On April 1, 2015, an investor has a portfolio consisting of eight securities as shown below:

Security	Market Price	No. of Shares	Value
A	29.40	400	0.59
B	318.70	800	1.32
C	660.20	150	0.87
D	5.20	300	0.35
E	281.90	400	1.16
F	275.40	750	1.24
G	514.60	300	1.05
H	170.50	900	0.76

The cost of capital for the investor is 20% p.a. continuously compounded. The investor fears a fall in the prices of the shares in the near future. Accordingly, he approaches you for the advice to protect the interest of his portfolio.

You can make use of the following information:

(1) The current NIFTY value is 8500.

(2) NIFTY futures can be traded in units of 25 only.

(3) Futures for May are currently quoted at 8700 and Futures for June are being quoted at 8850.

You are required to calculate:

(i) the beta of his portfolio.

(ii) the theoretical value of the futures contract for contracts expiring in May and June.

Given ($e^{0.03}$ =1.03045, $e^{0.04}$ = 1.04081, $e^{0.05}$ =1.05127)

(iii) the number of NIFTY contracts that he would have to sell if he desires to hedge until June in each of the following cases:

(A) His total portfolio

(B) 50% of his portfolio

(C) 120% of his portfolio

Solution — ICAI

13. (i) Beta of the Portfolio

Security	Market Price	No. of Shares	Value	β	Value x β
A	29.40	400	11760	0.59	6938.40
B	318.70	800	254960	1.32	336547.20
C	660.20	150	99030	0.87	86156.10
D	5.20	300	1560	0.35	546.00
E	281.90	400	112760	1.16	130801.60
F	275.40	750	206550	1.24	256122.00
G	514.60	300	154380	1.05	162099.00
H	170.50	900	153450	0.76	116622.00
			994450		1095832.30

$$\text{Portfolio Beta} = \frac{10,95,832.30}{9,94,450} = 1.102$$

(ii) Theoretical Value of Future Contract Expiring in May and June

$F = Se^{rt}$

$F_{May} = 8500 \times e^{0.20 \times (2/12)} = 8500 \times e^{0.0333}$

$e^{0.0333}$ shall be computed using Interpolation Formula as follows:

$e^{0.03}$	= 1.03045
$e^{0.04}$	= 1.04081
$e^{0.01}$	= 0.01036
$e^{0.0033}$	= 0.00342
$e^{0.0067}$	= 0.00694

$e^{0.0333}$ = 1.03045 + 0.00342 = 1.03387 or 1.04081 – 0.00694 = 1.03387

According the price of the May Contract

8500 X 1.03387 = ₹ 8788

Price of the June Contract

$F_{May} = 8500 \times e^{0.20 \times (3/12)} = 8500 \times e^{0.05} = 8500 \times 1.05127 = 8935.80$

(iii) No. of NIFTY Contracts required to sell to hedge until June

$$= \frac{\text{Value of Position to be hedged}}{\text{Value of Future Contract}} \times \beta$$

(A) Total portfolio

$$\frac{994450}{8850 \times 25} \times 1.102 = 4.953 \text{ say } 5 \text{ contracts}$$

(B) 50% of Portfolio

$$\frac{994450 \times 0.50}{8850 \times 25} \times 1.102 = 2.47 \text{ say } 3 \text{ contracts}$$

(C) 120% of Portfolio

$$\frac{994450 \times 1.20}{8850 \times 25} \times 1.102 = 5.94 \text{ say } 6 \text{ contracts}$$

Practical Q14

14. Sensex futures are traded at a multiple of 50. Consider the following quotations of Sensex futures in the 10 trading days during February, 2009:

Day	High	Low	Closing
4-2-09	3306.4	3290.00	3296.50
5-2-09	3298.00	3262.50	3294.40
6-2-09	3256.20	3227.00	3230.40
7-2-09	3233.00	3201.50	3212.30
10-2-09	3281.50	3256.00	3267.50
11-2-09	3283.50	3260.00	3263.80
12-2-09	3315.00	3286.30	3292.00
14-2-09	3315.00	3257.10	3309.30
17-2-09	3278.00	3249.50	3257.80
18-2-09	3118.00	3091.40	3102.60

Abshishek bought one sensex futures contract on February, 04. The average daily absolute change in the value of contract is ₹ 10,000 and standard deviation of these changes is ₹ 2,000. The maintenance margin is 75% of initial margin.

You are required to determine the daily balances in the margin account and payment on margin calls, if any.

Solution — ICAI

14. Initial Margin = $\mu + 3\sigma$

Where μ = Daily Absolute Change

σ = Standard Deviation

Accordingly

Initial Margin = ₹ 10,000 + ₹ 6,000 = ₹ 16,000

Maintenance margin = ₹ 16,000 x 0.75 = ₹ 12,000

Day	Changes in future Values (₹)	Margin A/c (₹)	Call Money (₹)
4/2/09	-	16000	-
5/2/09	50 x (3294.40 - 3296.50) = -105	15895	-
6/2/09	50 x (3230.40 - 3294.40) = -3200	12695	-
7/2/09	50 x (3212.30 - 3230.40) = -905	16000	4210
10/2/09	50 x (3267.50 - 3212.30) = 2760	18760	-
11/2/09	50 x (3263.80 - 3267.50) = -185	18575	-
12/2/09	50 x (3292 - 3263.80) =1410	19985	-
14/2/09	50 x (3309.30 - 3292) = 865	20850	-
17/2/09	50 x (3257.80 - 3309.30) = -2575	18275	-
18/2/09	50 x (3102.60 - 3257.80) = -7760	16000	5485

Practical Q15

15. Mr. A purchased a 3 month call option for 100 shares in XYZ Ltd. at a premium of ₹ 30 per share, with an exercise price of ₹ 550. He also purchased a 3 month put option for 100 shares of the same company at a premium of ₹ 5 per share with an exercise price of ₹ 450. The market price of the share on the date of Mr. A's purchase of options, is ₹ 500. Calculate the profit or loss that Mr. A would make assuming that the market price falls to ₹ 350 at the end of 3 months.

Solution — ICAI

15. Since the market price at the end of 3 months falls to ₹ 350 which is below the exercise price under the call option, the call option will not be exercised. Only put option becomes viable.

	₹
The gain will be:	
Gain per share (₹450 – ₹ 350)	100
Total gain per 100 shares	10,000
Cost or premium paid (₹ 30 x 100) + (₹ 5 x 100)	3,500
Net gain	6,500

Practical Q16

16. The market received rumour about ABC corporation's tie-up with a multinational company. This has induced the market price to move up. If the rumour is false, the ABC corporation stock price will probably fall dramatically. To protect from this an investor has bought the call and put options.

He purchased one 3 months call with a striking price of ₹ 42 for ₹ 2 premium, and paid Re.1 per share premium for a 3 months put with a striking price of ₹ 40.

(i) Determine the Investor's position if the tie up offer bids the price of ABC Corporation's stock up to ₹ 43 in 3 months.

(ii) Determine the Investor's ending position, if the tie up programme fails and the price of the stocks falls to ₹ 36 in 3 months.

Solution — ICAI

16. Cost of Call and Put Options

= (₹ 2 per share) x (100 share call) + (₹ 1 per share) x (100 share put)

= ₹ 2 x 100 + 1 x 100

= ₹ 300

(i) Price increases to ₹43. Since the market price is higher than the strike price of the call, the investor will exercise it.

Ending position = (- ₹ 300 cost of 2 option) + (₹ 1 per share gain on call) x 100

= - ₹ 300 + 100

Net Loss = - ₹ 200

(ii) The price of the stock falls to ₹36. Since the market price is lower than the strike price, the investor may not exercise the call option.

Ending Position = (- ₹300 cost of 2 options) + (₹4 per stock gain on put) x 100

= - ₹300 + 400

Gain = ₹100

Practical Q17

17. Equity share of PQR Ltd. is presently quoted at ₹ 320. The Market Price of the share after 6 months has the following probability distribution:

Market Price	₹ 180	260	280	320	400
Probability	0.1	0.2	0.5	0.1	0.1

A put option with a strike price of ₹ 300 can be written.

You are required to find out expected value of option at maturity (i.e. 6 months)

Solution — ICAI

17. Expected Value of Option

(300 – 180) X 0.1	12
(300 – 260) X 0.2	8
(300 – 280) X 0.5	10
(300 – 320) X 0.1	Not Exercised*
(300 – 400) X 0.1	Not Exercised*
	30

* If the strike price goes beyond ₹ 300, option is not exercised at all.

In case of Put option, since Share price is greater than strike price Option Value would be zero.

Practical Q18

18. You as an investor had purchased a 4 month call option on the equity shares of X Ltd. of ₹ 10, of which the current market price is ₹ 132 and the exercise price ₹ 150. You expect the price to range between ₹ 120 to ₹ 190. The expected share price of X Ltd. and related probability is given below:

Expected Price (₹)	120	140	160	180	190
Probability	0.05	0.20	0.50	0.10	0.15

Compute the following:

(i) Expected Share price at the end of 4 months.

(ii) Value of Call Option at the end of 4 months, if the exercise price prevails.

(iii) In case the option is held to its maturity, what will be the expected value of the call option?

Solution — ICAI

18. (i) Expected Share Price

= ₹120X 0.05 + ₹140X 0.20 + ₹160X 0.50 + ₹180X 0.10 + ₹190X 0.15

= ₹6 + ₹28 + ₹80 + ₹18 + ₹28.50 = ₹160.50

(ii) Value of Call Option

= ₹150 - ₹150 = Nil

(iii) If the option is held till maturity the expected Value of Call Option

Expected price (X)	Value of call (C)	Probability (P)	CP
₹ 120	0	0.05	0
₹ 140	0	0.20	0
₹ 160	₹ 10	0.50	₹ 5
₹ 180	₹ 30	0.10	₹ 3
₹ 190	₹ 40	0.15	₹ 6
Total			₹ 14

Alternatively, it can also be calculated as follows:

Expected Value of Option

(120 – 150) X 0.1	Not Exercised*
(140 – 150) X 0.2	Not Exercised*
(160 – 150) X 0.5	5
(180 – 150) X 0.1	3
(190 – 150) X 0.15	6
	14

* If the stock price goes below ₹ 150, option is not exercised at all.

Practical Q19

19. Mr. X established the following strategy on the Delta Corporation's stock :

(1) Purchased one 3-month call option with a premium of ₹ 30 and an exercise price of ₹ 550.

(2) Purchased one 3-month put option with a premium of ₹ 5 and an exercise price of ₹ 450.

Delta Corporation's stock is currently selling at ₹ 500. Determine profit or loss, if the price of Delta Corporation's stock:

(i) remains at ₹ 500 after 3 months.

(ii) falls at ₹ 350 after 3 months.

(iii) rises to ₹ 600.

Assume the option size is 100 shares of Delta Corporation.

Solution — ICAI

19. Total premium paid on purchasing a call and put option

= (₹30 per share × 100) + (₹5 per share × 100).

= 3,000 + 500 = ₹3,500

(i) In this case, X exercises neither the call option nor the put option as both will result in a loss for him.

Ending value = - ₹3,500 + zero gain = - ₹3,500

i.e Net loss = ₹3,500

(ii) Since the price of the stock is below the exercise price of the call, the call will not be exercised. Only put is valuable and is exercised.

Total premium paid = ₹3,500

Ending value = – ₹3,500 + ₹[(450 – 350) × 100] = – ₹3,500 + ₹10,000 = ₹6,500

∴ Net gain = ₹6,500

(iii) In this situation, the put is worthless, since the price of the stock exceeds the put's exercise price. Only call option is valuable and is exercised.

Total premium paid = ₹3,500

Ending value = -3,500 +[(600 – 550) × 100]

Net Gain = -3,500 + 5,000 = ₹1,500

Practical Q20

20. The equity share of VCC Ltd. is quoted at ₹ 210. A 3-month call option is available at a premium of ₹ 6 per share and a 3-month put option is available at a premium of ₹ 5 per share. Ascertain the net payoffs to the option holder of a call option and a put option separately.

(i) the strike price in both cases in ₹ 220; and

(ii) the share price on the exercise day is ₹ 200,210,220,230,240.

Also indicate the price range at which the call and the put options may be gainfully exercised.

Solution — ICAI

20. Net payoff for the holder of the call option

					(₹)
Share price on exercise day	200	210	220	230	240
Option exercise	No	No	No	Yes	Yes
Outflow (Strike price)	Nil	Nil	Nil	220	220
Out flow (premium)	6	6	6	6	6
Total Outflow	6	6	6	226	226
Less inflow (Sales proceeds)	-	-	-	230	240
Net payoff	-6	-6	-6	4	14

Net payoff for the holder of the put option

					(₹)
Share price on exercise day	200	210	220	230	240
Option exercise	Yes	Yes	No	No	No
Inflow (strike price)	220	220	Nil	Nil	Nil
Less outflow (purchase price)	200	210	-	-	-
Less outflow (premium)	5	5	5	5	5
Net Payoff	15	5	-5	-5	-5

The call option can be exercised gainfully for any price above ₹226 (₹220 + ₹6) and put option for any price below ₹215 (₹220 - ₹5).

Practical Q21

21. Sumana wanted to buy shares of EIL which has a range of ₹ 411 to ₹ 592 a month later. The present price per share is ₹ 421. Her broker informs her that the price of this share can sore up to ₹ 522 within a month or so, so that she should buy a one-month CALL of EIL. In order to be prudent in buying the call, the share price should be more than or at least ₹ 522 the assurance of which could not be given by her broker.

Though she understands the uncertainty of the market, she wants to know the probability of attaining the share price ₹ 592 so that buying of a one-month CALL of EIL at the execution price of ₹ 522 is justified. Advice her. Take the risk-free interest to be 3.60% for one month and $e^{0.036} = 1.037$.

Solution — ICAI

21. $p = \frac{e^{rt} - d}{u - d}$

$e^{rt} = e^{0.036}$

d = 411/421 = 0.976

u = 592/421 = 1.406

$p = \frac{e^{0.036} - 0.976}{1.406 - 0.976} = \frac{1.037 - 0.976}{0.43} = \frac{0.061}{0.43} = 0.1418$

Thus probability of rise in price 0.1418

Practical Q22

22. Mr. Dayal is interested in purchasing equity shares of ABC Ltd. which are currently selling at ₹ 600 each. He expects that price of share may go upto ₹ 780 or may go down to ₹ 480 in three months. The chances of occurring such variations are 60% and 40% respectively. A call option on the shares of ABC Ltd. can be exercised at the end of three months with a strike price of ₹ 630.

(i) What combination of share and option should Mr. Dayal select if he wants a perfect hedge?

(ii) What should be the value of option today (the risk free rate is 10% p.a.)?

(iii) What is the expected rate of return on the option?

Solution — ICAI

22. (i) To compute perfect hedge we shall compute Hedge Ratio (Δ) as follows:

$$\Delta = \frac{C_1 - C_2}{S_1 - S_2} = \frac{150 - 0}{780 - 480} = \frac{150}{300} = 0.50$$

Mr. Dayal should purchase 0.50 share for every 1 call option.

(ii) Value of Option today

If price of share comes out to be ₹780 then value of purchased share will be:

Sale Proceeds of Investment (0.50 x ₹ 780)	₹ 390
Loss on account of Short Position (₹ 780 – ₹ 630)	₹ 150
	₹ 240

If price of share comes out to be ₹ 480 then value of purchased share will be:

Sale Proceeds of Investment (0.50 x ₹ 480)	₹ 240

Accordingly, Premium say P shall be computed as follows:

(₹ 300 – P) 1.025 = ₹ 240

P = ₹65.85

(iii) Expected Return on the Option

Expected Option Value = (₹ 780 – ₹ 630) × 0.60 + ₹ 0 × 0.40 = ₹ 90

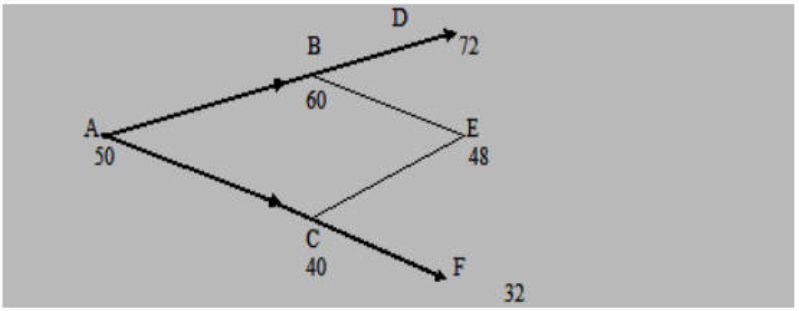
$$\text{Expected Rate of Return} = \frac{90 - 65.85}{65.85} \times 100 = 36.67\%$$

Practical Q23

23. Consider a two-year call option with a strike price of ₹ 50 on a stock the current price of which is also ₹ 50. Assume that there are two-time periods of one year and in each year the stock price can move up or down by equal percentage of 20%. The risk-free interest rate is 6%. Using binominal option model, calculate the probability of price moving up and down. Also draw a two-step binomial tree showing prices and payoffs at each node.

Solution — ICAI

23. Stock prices in the two step Binominal tree

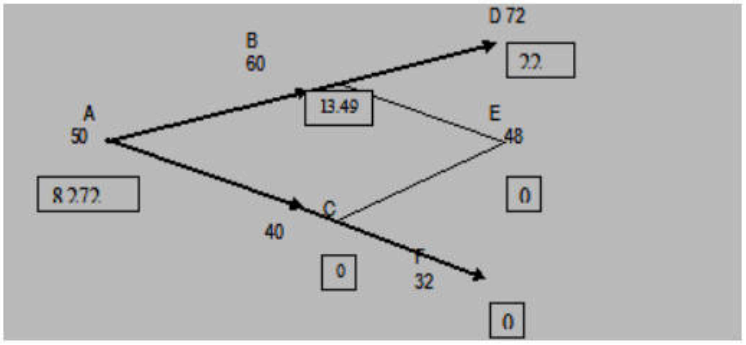


Using the single period model, the probability of price increase is

$$P = \frac{R-d}{u-d} = \frac{1.06-0.80}{1.20-0.80} = \frac{0.26}{0.40} = 0.65$$

therefore the p of price decrease = 1-0.65 = 0.35

The two step Binominal tree showing price and pay off



The value of an American call option at nodes D, E and F will be equal to the value of European option at these nodes and accordingly the call values at nodes D, E and F will be 22, 0 and 0 using the single period binomial model the value of call option at node B is

$$C = \frac{Cup+Cd(1-p)}{R} = \frac{22\times0.65+0\times0.35}{1.06} = 13.49$$

The value of option at node 'A' is

$$\frac{13.49\times0.65+0\times0.35}{1.06} = 8.272$$

Practical Q24

24. The current market price of an equity share of Penchant Ltd is ₹ 420. Within a period of 3 months, the maximum and minimum price of it is expected to be ₹ 500 and ₹ 400 respectively. If the risk free rate of interest be 8% p.a., what should be the value of a 3 months Call option

under the "Risk Neutral" method at the strike rate of ₹ 450?

Given $e^{0.02} = 1.0202$

Solution — ICAI

24. Let the probability of attaining the maximum price be p

(500 - 420) x p+(400 - 420) x (1-p) = 420 x ($e^{0.02}$.1)

or, 80p - 20(1 - p) = 420 x 0.0202

or, 80p – 20 + 20p = 8.48

or, 100p = 28.48

p = 0.2848

The value of Call Option in ₹ $= \frac{0.2848x(500-450)}{1.0202} = \frac{0.2848x50+0.7152\times0}{1.0202} = 13.96$

Practical Q25

25. From the following data for certain stock, find the value of a call option:

Price of stock now	=	₹ 80
Exercise price	=	₹ 75
Standard deviation of continuously compounded annual return	=	0.40
Maturity period	=	6 months
Annual interest rate	=	12%

Given

Number of S.D. from Mean, (z)	Area of the left or right (one tail)
0.25	0.4013
0.30	0.3821
0.55	0.2912
0.60	0.2743

$e^{0.12x0.5}$ = 1.062

In 1.0667 = 0.0646

Solution — ICAI

25. Applying the Black Scholes Formula,

Value of the Call option now:

The Formula $C = SN(d_1) - Ke^{(-rt)} N(d_2)$

$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$

$d_2 = d_1 - \sigma\sqrt{t}$

Where,

C = Theoretical call premium

S = Current stock price

t = time until option expiration

K = option striking price

r = risk-free interest rate

N = Cumulative standard normal distribution

e = exponential term

σ = Standard deviation of continuously compounded annual return.

ln = natural logarithim

$$d_1 = \frac{\ln(1.0667) + (12\% + 0.08)0.5}{0.40\sqrt{0.5}}$$

$$= \frac{0.0646 + (0.2)0.5}{0.40 \times 0.7071} = \frac{0.1646}{0.2828} = 0.5820$$

d_2 = 0.5820 – 0.2828 = 0.2992

$N(d_1)$ = N (0.5820)

$N(d_2)$ = N (0.2992)

Price = $SN(d_1) - Ke^{(-rt)}N(d_2)$ = 80 x $N(d_1)$ – (75/1.062) x $N(d_2)$

Value of option

$$= 80\ N(d_1) - \frac{75}{1.062} \times N(d_2)$$

$N(d_1)$ = N (0.5820) = 0.7197

$N(d_2)$ = N(0.2992) = 0.6176

$$\text{Price} = 80 \times 0.7197 - \frac{75}{1.062} \times 0.6176$$

= 57.57 – 70.62 x 0.6176 = 57.57 – 43.61 = ₹13.96

Teaching Notes:

Students may please note following important point:

Values of $N(d_1)$ and $N(d_2)$ have been computed by interpolating the values of areas under respective

TYK practical with ICAI solutions

numbers of SD from Mean (Z) given in the question.

It may also be possible that in question paper areas under Z may be mentioned otherwise e.g. Cumulative Area or Area under Two tails. In such situation the areas of the respective Zs given in the question will be as follows:

Cumulative Area

Number of S.D. from Mean, (z)	*Cumulative Area*
0.25	0.5987
0.30	0.6179
0.55	0.7088
0.60	0.7257

Two tail area

Number of S.D. from Mean, (z)	*Area of the left and right (two tail)*
0.25	0.8026
0.30	0.7642
0.55	0.5823
0.60	0.5485

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